



Lecture 3.1: Performance of Transmission Lines

EEE 321: Power System I

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Presentation Outline

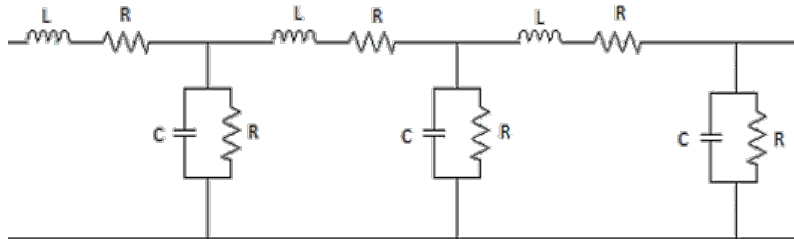
- ❑ Classification of Overhead Transmission Lines
- ❑ Short Transmission Lines
- ❑ Medium Transmission Lines
- ❑ Long Transmission Lines
- ❑ Transmission Line as Two-port network
- ❑ Problems and solutions

Classification of Overhead Transmission Lines

TRANSMISSION LINE

- Short transmission line (STL)
Lines less than 50 km
Voltage level: up to 20KV
- Medium transmission line (MTL)
Lines between 50 and 150 km
Voltage level: 20KV to 100KV
- Long transmission line (LTL)
Lines longer than 150 km
Voltage level: above 100KV

Circuit Parameters of Overhead Transmission Lines



An equivalent circuit of line can be represented by four electrical parameters:

Series resistance (R)

Depends on the physical composition of the conductor at a given temperature and causes actual power loss in the conductor. It is important for transmission efficiency evaluation and economic studies.

Series inductance (L)

Due to the magnetic properties of the conductor. Power transmission capacity of T. L is basically governed by inductance.

Shunt conductance (G)

Due to the leakage current flowing across insulators. The leakage currents are negligible compared to the transmission line current.

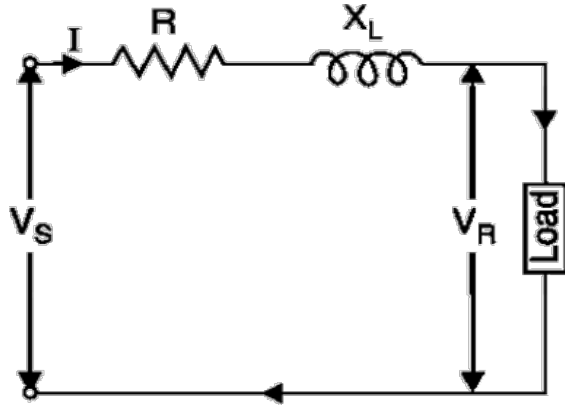
Shunt capacitance (C)

Due to the electrical field properties between conductors

Electrical performances of Transmission Line i.e. efficiency and voltage regulation depend on the value of line parameters. Therefore, for designing a line, knowledge of calculating and setting the values of these parameters is very important.

Equivalent Circuit of Single Phase STL

In short transmission line performance analysis,



V_s = Sending end voltage

I = Load current

R = Loop resistance i.e., resistance of both conductors

X_L = Loop inductive reactance of the short transmission line

V_R = Receiving end voltage

Load is inductive load (lagging load)

Equivalent circuit of a single phase STL

Assumptions:

1. Shunt conductance is assumed zero (as it is considered in the STL that there is no leakage current flowing from conductor to earth).
2. Shunt capacitance effect is assumed zero. Since capacitors are connected in parallel and its total value is proportional to the line length. Therefore when the length of the line is short, there will be less shunt capacitance. So, we can ignore the shunt capacitance value and capacitance dependent parameter calculations in the STL. For example: charging current will be less as it depends on the capacitance and voltage rating. Here for short transmission line-both capacitance and voltage are less, therefore, we can neglect the inclusion of charging current in the STL calculations.

Single phase STL has two conductors. Each conductor has resistance and inductive reactance. For convenience, it is considered that the parameters of both conductors i.e. resistance and reactance are lumped into one conductor, and assume the return conductor has no resistance and reactance.

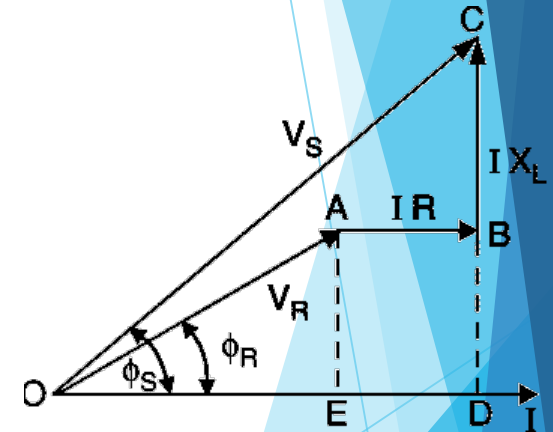
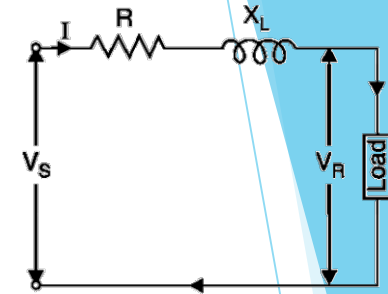
Phasor diagram of Single Phase STL

From equivalent circuit, we find

$$V_S = IR + IX_L + V_R$$

Drawing phasor diagram:

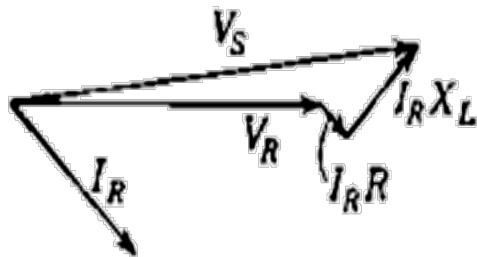
1. Current I is taken as the reference phasor
2. Since the load is inductive, consider receiving end voltage V_R voltage leads current I by ϕ_R . In phasor diagram, OA represents the V_R
3. IR is the resistive voltage drop. Therefore, this voltage drop is parallel to the load current I . Here, AB represents the IR
4. IX_L is the inductive voltage drop. Therefore, it leads I by 90° . Here BC represents the IX_L
5. Finally the sum of V_R , IR and IX_L voltages are drawn as V_S . This voltage V_S leads I by ϕ_S . So here OC represents the V_S



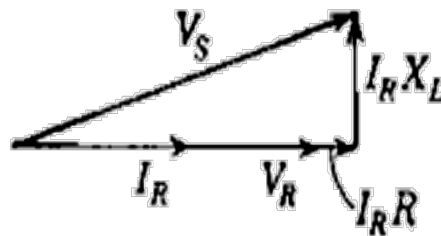
Phasor diagram of a STL



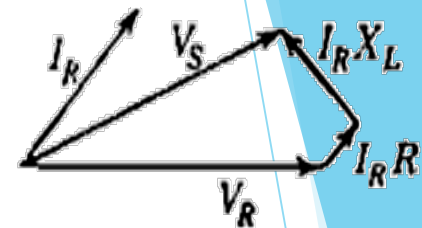
Phasor diagram of Single Phase STL



(a) Load p. f. = 70% lag



(b) Load p. f. = 100%



(c) Load p. f. = 70% lead

Phasor diagrams of a short transmission line. All diagrams are drawn for the same magnitudes of V_R and I_R .

Phasor diagram was drawn using current I as reference in the previous slide, here V_R is taken as reference



Performance of STL

or

$$(OC)^2 = (OD)^2 + (DC)^2$$

$$V_S^2 = (OE + ED)^2 + (DB + BC)^2$$

$$= (V_R \cos \phi_R + IR)^2 + (V_R \sin \phi_R + IX_L)^2$$

$\therefore$

$$V_S = \sqrt{(V_R \cos \phi_R + IR)^2 + (V_R \sin \phi_R + IX_L)^2}$$

(i) %age Voltage regulation = $\frac{V_S - V_R}{V_R} \times 100$

(ii) Sending end *p.f.*, $\cos \phi_S = \frac{OD}{OC} = \frac{V_R \cos \phi_R + IR}{V_S}$

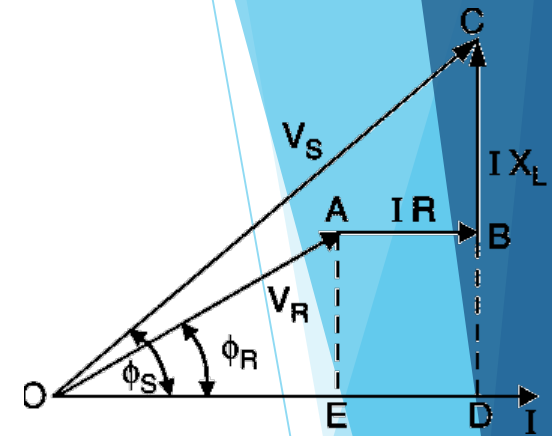
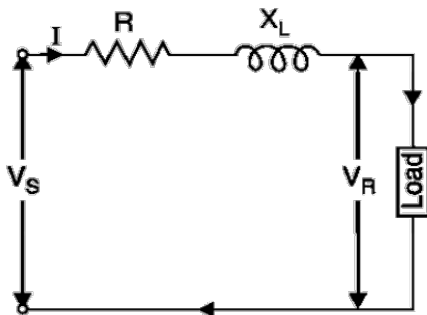
(iii) Power delivered = $V_R I_R \cos \phi_R$

Line losses = $I^2 R$

Power sent out = $V_R I_R \cos \phi_R + I^2 R$

%age Transmission efficiency = $\frac{\text{Power delivered}}{\text{Power sent out}} \times 100$

$$= \frac{V_R I_R \cos \phi_R}{V_R I_R \cos \phi_R + I^2 R} \times 100$$



* $I_R = I_S = I$

$\frac{OE}{OA} = \cos \phi_R$

$\therefore OE = OA \cos \phi_R = V_R \cos \phi_R$

$\frac{EA}{OA} = \sin \phi_R$

$\therefore EA = OA \sin \phi_R = V_R \sin \phi_R$

$\therefore DB = V_R \sin \phi_R$ (As $DB = EA$)

Problems on STL

Problem 1. A single phase overhead transmission line delivers 1100 kW at 33 kV at 0.8 p.f. lagging. The total resistance and inductive reactance of the line are 10Ω and 15Ω respectively. Determine : (i) sending end voltage (ii) sending end power factor and (iii) transmission efficiency.

Solution: Given

Resistance = 10Ω

Reactance = 15Ω

Power delivered = 1100 kW

Receiving end voltage, $V_R = 33 \text{ kV} = 33,000 \text{ V}$

Receiving end power factor, $\cos \phi_R = 0.8$ lagging

Total line impedance, $Z = R + jX_L = 10 + j15$

(i) sending end voltage,

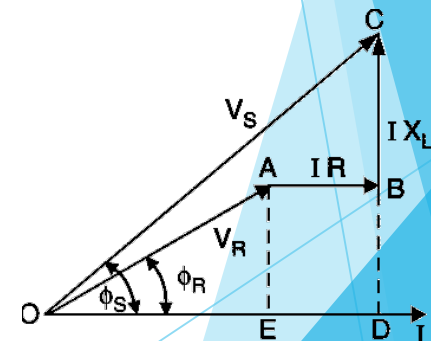
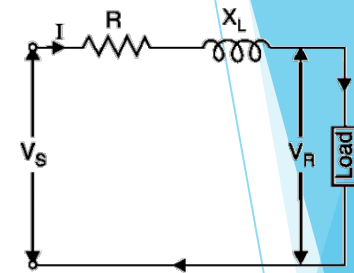
$$V_S = \sqrt{(V_R \cos \phi_R + IR)^2 + (V_R \sin \phi_R + IX_L)^2}$$

(ii) sending end power factor, $\cos \phi_S$

$$= \frac{V_R \cos \phi_R + IR}{V_S}$$

(iii) Transmission efficiency

$$= \frac{V_R I_R \cos \phi_R}{V_R I_R \cos \phi_R + I^2 R} \times 100$$



Problems on Single Phase STL

(i) sending end voltage, $V_S = \sqrt{(V_R \cos \phi_R + IR)^2 + (V_R \sin \phi_R + IX_L)^2}$ (1)

$$\text{Power delivered} = V_R I_R \cos \phi_R$$

$$I_R = \frac{\text{Power delivered}}{V_R \cos \phi_R} = 41.67 \text{ A}$$

$$\therefore I_R = I = 41.67 \text{ A}$$

$$\cos \phi_R = 0.8 \quad \therefore \sin \phi_R = 0.6$$

Now from eqn. (1) we get sending end voltage, $V_S = 33709 \text{ V}$

(ii) sending end power factor, $\cos \phi_S = \frac{V_R \cos \phi_R + IR}{V_S} = 0.7955 \text{ lagging}$

(iii) Transmission efficiency = $\frac{V_R I_R \cos \phi_R}{V_R I_R \cos \phi_R + I^2 R} \times 100 = 98.44\%$

Given

Resistance = 10Ω

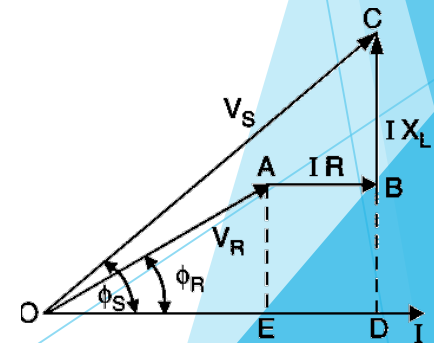
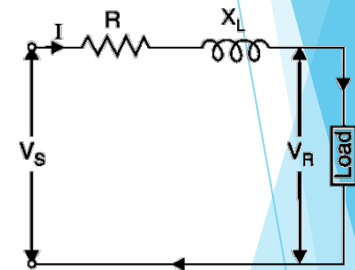
Reactance = 15Ω

Power delivered = 1100 kW

Receiving end voltage, $V_R = 33 \text{ kV} = 33,000 \text{ V}$

Receiving end power factor, $\cos \phi_R = 0.8 \text{ lagging}$

Total line impedance, $Z = R + jX_L = 10 + j15$



Equivalent Circuit of Three Phase STL

For reasons associated with economy, transmission of electric power is done by 3-phase system. This system may be regarded as consisting of three single phase units, each wire transmitting one-third of the total power. As a matter of convenience, we generally analyse 3-phase system by considering

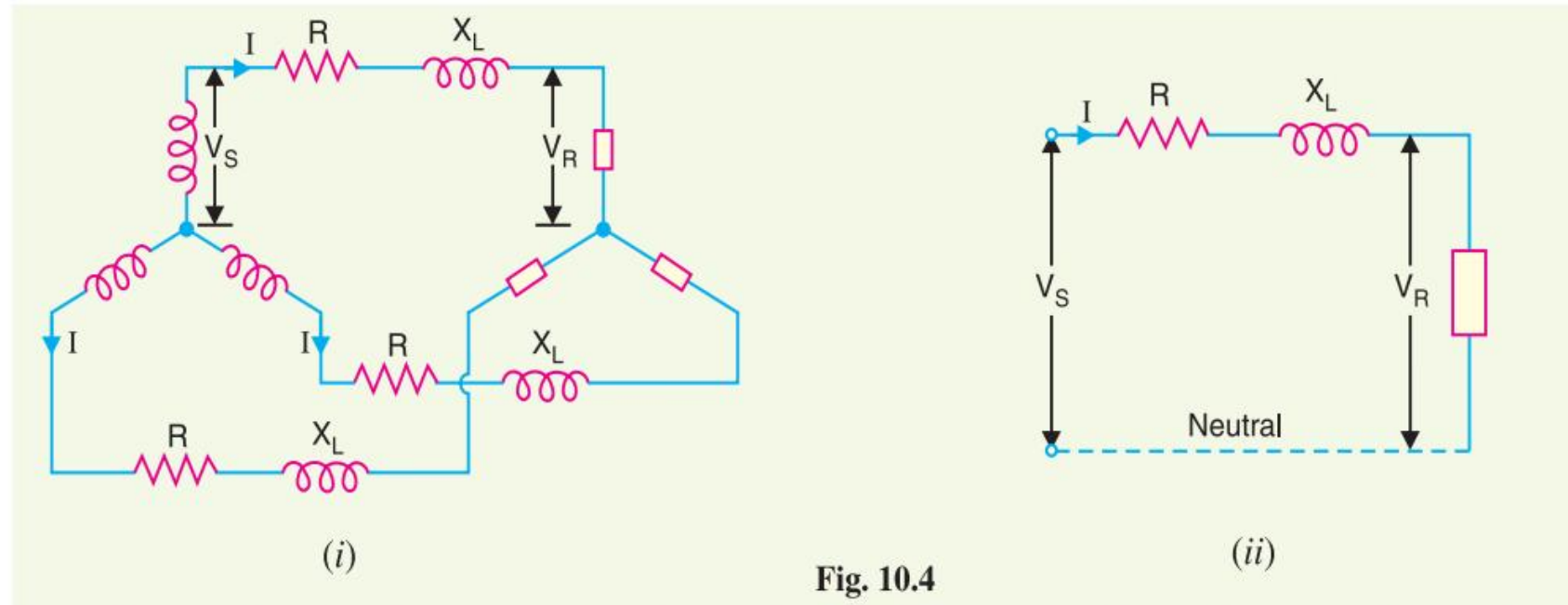


Fig. 10.4

*one phase only. Therefore, expression for regulation, efficiency etc. derived for a single phase line can also be applied to a 3-phase system. Since only one phase is considered, phase values of 3-phase system should be taken. Thus, V_S and V_R are the phase voltages, whereas R and X_L are the resistance and inductive reactance per phase respectively.

Fig. 10.4 (i) shows a Y-connected generator supplying a balanced Y-connected load through a transmission line. Each conductor has a resistance of $R \Omega$ and inductive reactance of $X_L \Omega$. Fig. 10.4 (ii) shows one phase separately. The calculations can now be made in the same way as for a single phase line.

Problems of Three Phase STL

Example 10.3. An overhead 3-phase transmission line delivers 5000 kW at 22 kV at 0.8 p.f. lagging. The resistance and reactance of each conductor is 4 Ω and 6 Ω respectively. Determine : (i) sending end voltage (ii) percentage regulation (iii) transmission efficiency.

Solution.

Load power factor, $\cos \phi_R = 0.8$ lagging

Receiving end voltage/phase, $*V_R = 22,000/\sqrt{3} = 12,700$ V

Impedance/phase, $\vec{Z} = 4 + j 6$

Line current, $I = \frac{5000 \times 10^3}{3 \times 12700 \times 0.8} = 164$ A

As $\cos \phi_R = 0.8 \quad \therefore \sin \phi_R = 0.6$

Taking $\vec{V}_R$ as the reference phasor (see Fig. 10.6),

$$\vec{V}_R = V_R + j 0 = 12700 \text{ V}$$

$$\vec{I} = I (\cos \phi_R - j \sin \phi_R) = 164 (0.8 - j 0.6) = 131.2 - j 98.4$$

(i) Sending end voltage per phase is

$$\begin{aligned} \vec{V}_S &= \vec{V}_R + \vec{I} \vec{Z} = 12700 + (131.2 - j 98.4) (4 + j 6) \\ &= 12700 + 524.8 + j 787.2 - j 393.6 + 590.4 \\ &= 13815.2 + j 393.6 \end{aligned}$$

$$\text{Magnitude of } V_S = \sqrt{(13815.2)^2 + (393.6)^2} = 13820.8 \text{ V}$$

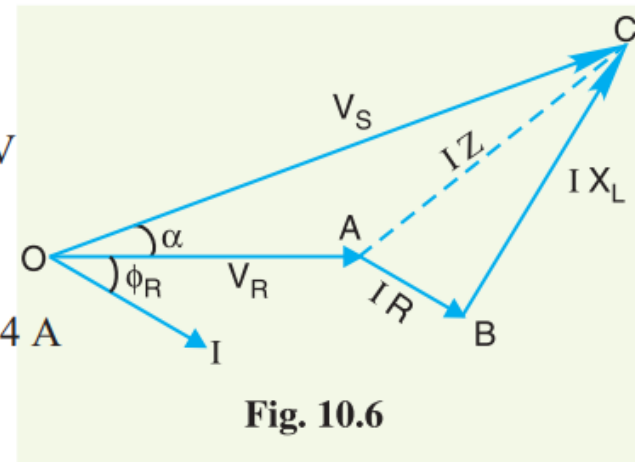


Fig. 10.6

* If not mentioned in the problem, star-connection is understood.

Problems of Three Phase STL

$$\text{Line value of } V_S = \sqrt{3} \times 13820.8 = 23938 \text{ V} = \mathbf{23.938 \text{ kV}}$$

$$(ii) \quad \% \text{ age Regulation} = \frac{V_S - V_R}{V_R} \times 100 = \frac{13820.8 - 12700}{12700} \times 100 = \mathbf{8.825\%}$$

$$(iii) \quad \text{Line losses} = 3I^2R = 3 \times (164)^2 \times 4 = 3,22,752 \text{ W} = 322.752 \text{ kW}$$

$$\therefore \text{Transmission efficiency} = \frac{5000}{5000 + 322.752} \times 100 = \mathbf{93.94\%}$$

Example 10.4. Estimate the distance over which a load of 15000 kW at a p.f. 0.8 lagging can be delivered by a 3-phase transmission line having conductors each of resistance 1 Ω per kilometre. The voltage at the receiving end is to be 132 kV and the loss in the transmission is to be 5%.

Solution.

$$\text{Line current, } I = \frac{\text{Power delivered}}{\sqrt{3} \times \text{line voltage} \times \text{power factor}} = \frac{15000 \times 10^3}{\sqrt{3} \times 132 \times 10^3 \times 0.8} = 82 \text{ A}$$

$$\text{Line losses} = 5\% \text{ of power delivered} = 0.05 \times 15000 = 750 \text{ kW}$$

Let $R \Omega$ be the resistance of one conductor.

$$\text{Line losses} = 3I^2R$$

$$\text{or } 750 \times 10^3 = 3 \times (82)^2 \times R$$

$$\therefore R = \frac{750 \times 10^3}{3 \times (82)^2} = 37.18 \Omega$$

Resistance of each conductor per km is 1 Ω (given).

$$\therefore \text{Length of line} = \mathbf{37.18 \text{ km}}$$

TRANSMISSION LINE AS TWO PORT NETWORK

In general, a transmission line may be viewed as a four terminal (two-port) network, as shown in the following figure.

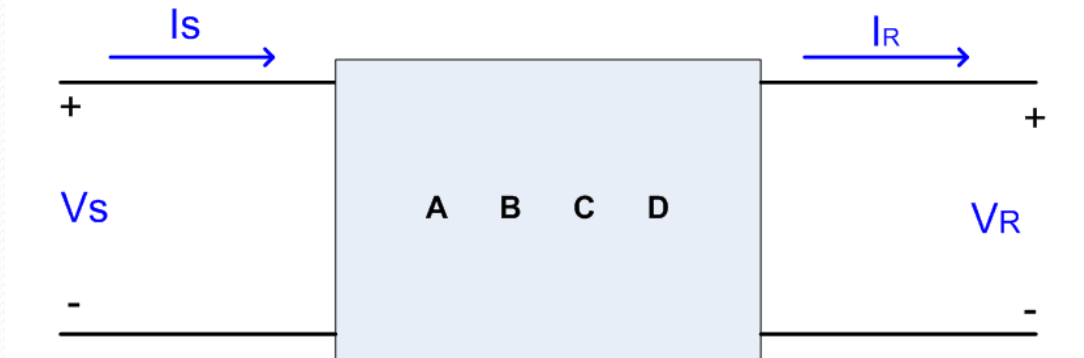


Figure: Two port network

In any four terminal *network, the input voltage and input current can be expressed in terms of output voltage and output current. Incidentally, a transmission line is a 4-terminal network ; two input terminals where power enters the network and two output terminals where power leaves the network.

Therefore, the input voltage ($\vec{V}_S$) and input current ($\vec{I}_S$) of a 3-phase transmission line can be expressed as :

$$\vec{V}_S = \vec{A} \vec{V}_R + \vec{B} \vec{I}_R$$

$$\vec{I}_S = \vec{C} \vec{V}_R + \vec{D} \vec{I}_R$$

where

$$\vec{V}_S = \text{sending end voltage per phase}$$

$$\vec{I}_S = \text{sending end current}$$

$$\vec{V}_R = \text{receiving end voltage per phase}$$

$$\vec{I}_R = \text{receiving end current}$$

TRANSMISSION LINE AS TWO PORT NETWORK

In general, a transmission line may be viewed as a four terminal (two-port) network, as shown in the following figure.

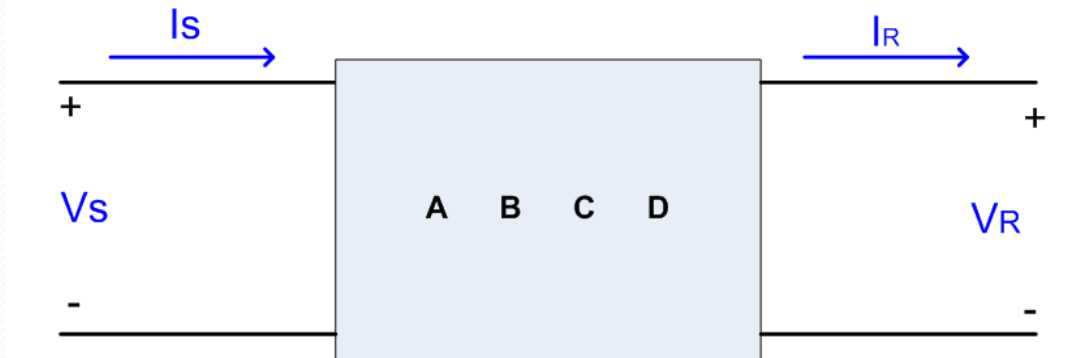


Figure: Two port network

and $\vec{A}$, $\vec{B}$, $\vec{C}$ and $\vec{D}$ (generally complex numbers) are the constants known as *generalised circuit constants* of the transmission line. The values of these constants depend upon the particular method adopted for solving a transmission line. Once the values of these constants are known, performance calculations of the line can be easily worked out. The following points may be kept in mind :

- (i) The constants $\vec{A}$, $\vec{B}$, $\vec{C}$ and $\vec{D}$ are generally complex numbers.
- (ii) The constants $\vec{A}$ and $\vec{D}$ are dimensionless whereas the dimensions of $\vec{B}$ and $\vec{C}$ are ohms and siemen respectively.

- (iii) For a given transmission line,

$$\vec{A} = \vec{D}$$

- (iv) For a given transmission line,

$$\vec{A} \vec{D} - \vec{B} \vec{C} = 1$$

We shall establish the correctness of above characteristics of generalised circuit constants in the following discussion.

Equivalent circuit of MTL

Assumptions:

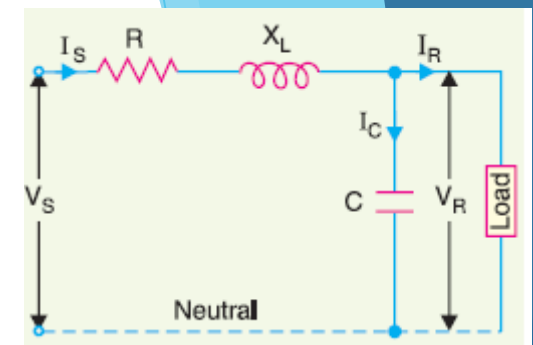
1. Shunt conductance is assumed zero (Ideal condition: no leakage current).
2. Shunt capacitance effect is not assumed zero. As capacitors are connected parallel and its total value is proportional to the line length. Therefore when the length of the line is medium, there will be some considerable shunt capacitance. As a result, the parameter in performance calculation which depends on the capacitance value, will also have some considerable value.

In transmission line, the capacitance is uniformly distributed over the entire length of the line. However, in order to make the calculations simple, **the line capacitance is assumed to be lumped or concentrated across the line at one or more points**. This localizing of the line capacitance gives reasonably accurate results. For the performance analysis of medium transmissions lines, commonly used localized capacitance models are:

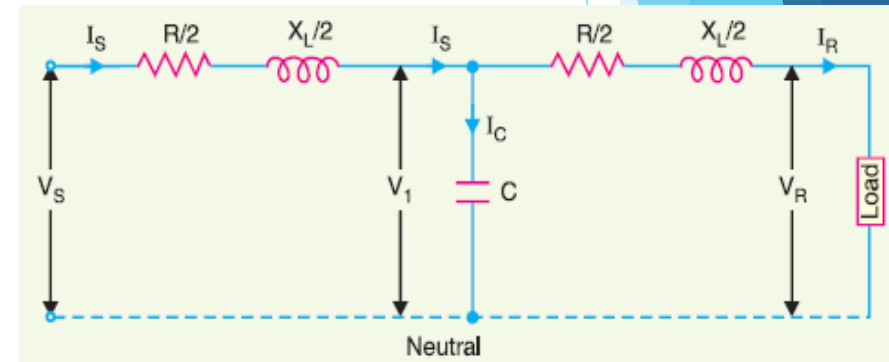
- (i) End Condenser model
- (ii) Nominal T model
- (iii) Nominal π model

In nominal T model of TL, the whole line capacitance is assumed to be concentrated at the middle point of the line and half the line resistance and reactance are lumped on each side of the capacitance.

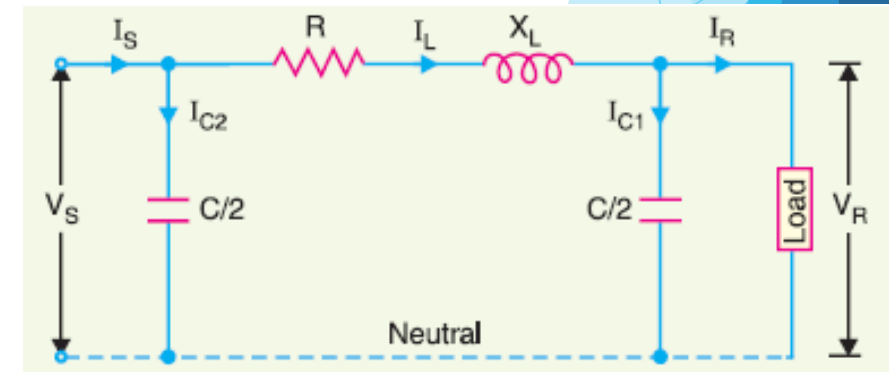
In nominal π model of TL, capacitance of each conductor (i.e., line to neutral) is divided into two halves; one half being lumped at the sending end and the other half at the receiving end.



End Condenser model



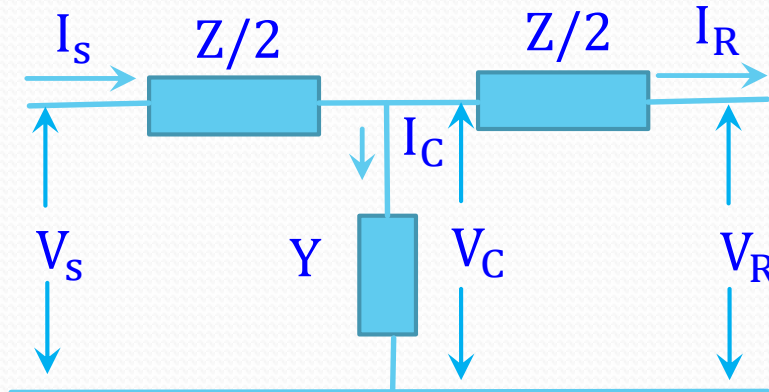
Nominal T model



Nominal π model

Equivalent circuit of MTL

MEDIUM TRANSMISSION LINE: NOMINAL T METHOD



$$V_C = V_R + I_R \frac{Z}{2};$$

$$I_C = V_C Y = \left(V_R + I_R \frac{Z}{2} \right) Y$$

$$I_S = I_R + I_C$$

$$I_S = I_R + \left(V_R + I_R \frac{Z}{2} \right) Y$$

$$I_S = V_R Y + \left(I_R + I_R Y \frac{Z}{2} \right)$$

$$I_S = Y V_R + \left(1 + \frac{YZ}{2} \right) I_R \quad (1)$$

$$V_S = V_C + I_S \frac{Z}{2}$$

$$V_S = V_R + I_R \frac{Z}{2} + \left[Y V_R + \left(1 + \frac{YZ}{2} \right) I_R \right] \frac{Z}{2}$$

$$V_S = V_R + I_R \frac{Z}{2} + V_R Y \frac{Z}{2} + I_R \frac{Z}{2} + I_R Y \frac{Z^2}{4}$$

$$V_S = \left(1 + \frac{YZ}{2} \right) V_R + \left(1 + \frac{YZ}{4} \right) Z I_R \quad (2)$$

Equivalent circuit of MTL

MEDIUM TRANSMISSION LINE: NOMINAL T METHOD

Equation (2) can be written as:

$$V_S = AV_R + BI_R$$

Equation (1) can be written as:

$$I_S = CV_R + DI_R$$

Where, $A = 1 + \frac{YZ}{2}$

$$B = \left(1 + \frac{YZ}{4}\right)Z$$

$$C = Y$$

$$D = 1 + \frac{YZ}{2}$$

$$AD - BC = \left(1 + \frac{YZ}{2}\right)\left(1 + \frac{YZ}{2}\right) - \left(1 + \frac{YZ}{4}\right)ZY = 1$$

Problems of MTL

Example 10.16. A balanced 3-phase load of 30 MW is supplied at 132 kV, 50 Hz and 0.85 p.f. lagging by means of a transmission line. The series impedance of a single conductor is $(20 + j52)$ ohms and the total phase-neutral admittance is 315×10^{-6} siemen. Using nominal T method, determine: (i) the A , B , C and D constants of the line (ii) sending end voltage (iii) regulation of the line.

Solution. Fig. 10.26 shows the representation of 3-phase line on the single phase basis.

Series line impedance/phase, $\vec{Z} = (20 + j 52) \Omega$

Shunt admittance/phase, $\vec{Y} = j 315 \times 10^{-6} \text{ S}$

(i) **Generalised constants of line.** For nominal T method, various constants have the values as under :

$$\begin{aligned}\vec{A} = \vec{D} &= 1 + \vec{Z} \vec{Y} / 2 = 1 + \frac{20 + j 52}{2} \times j 315 \times 10^{-6} \\ &= 0.992 + j 0.00315 = \mathbf{0.992 \angle 0.18^\circ}\end{aligned}$$

$$\begin{aligned}\vec{B} &= \vec{Z} \left(1 + \frac{\vec{Z} \vec{Y}}{4} \right) = (20 + j 52) \left[1 + \frac{(20 + j 52) j 315 \times 10^{-6}}{4} \right] \\ &= 19.84 + j 51.82 = \mathbf{55.5 \angle 69^\circ}\end{aligned}$$

$$\vec{C} = \vec{Y} = \mathbf{0.000315 \angle 90^\circ}$$

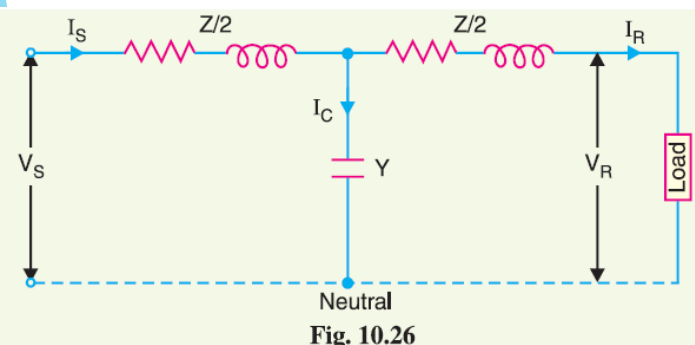


Fig. 10.26

Problems of MTL

(ii) Sending end voltage.

Receiving end voltage/phase, $V_R = 132 \times 10^3 / \sqrt{3} = 76210 \text{ V}$

Receiving end current, $I_R = \frac{30 \times 10^6}{\sqrt{3} \times 132 \times 10^3 \times 0.85} = 154 \text{ A}$

$$\cos \phi_R = 0.85 ; \quad \sin \phi_R = 0.53$$

Taking receiving end voltage as the reference phasor, we have,

$$\vec{V}_R = V_R + j0 = 76210 \text{ V}$$

$$\vec{I}_R = I_R (\cos \phi_R - j \sin \phi_R) = 154 (0.85 - j 0.53) = 131 - j 81.62$$

Sending end voltage per phase is

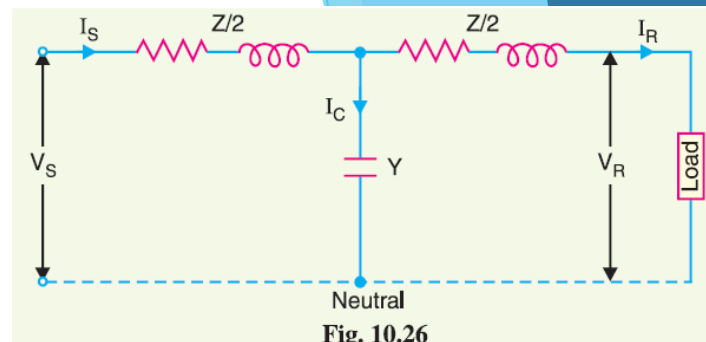
$$\begin{aligned} \vec{V}_S &= \vec{A} \vec{V}_R + \vec{B} \vec{I}_R \\ &= (0.992 + j 0.0032) 76210 + (19.84 + j 51.82) (131 - j 81.62) \\ &= 82,428 + j 5413 \end{aligned}$$

∴ Magnitude of sending end voltage is

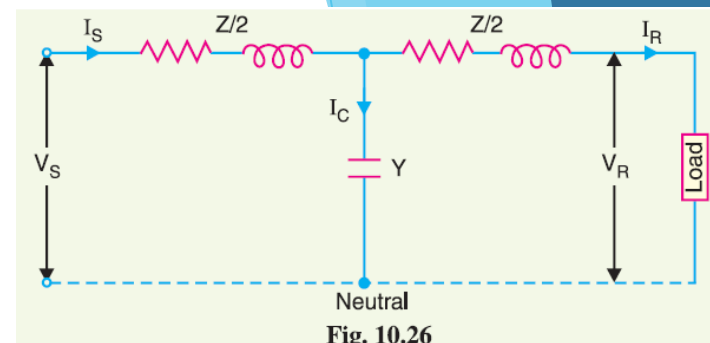
$$V_S = \sqrt{(82,428)^2 + (5413)^2} = 82.6 \times 10^3 \text{ V} = 82.6 \text{ kV}$$

∴ Sending end line-to-line voltage

$$= 82.6 \times \sqrt{3} = \mathbf{143 \text{ kV}}$$



Problems of MTL



(iii) Regulation. Regulation is defined as the change in voltage at the receiving end when full-load is thrown off.

Now,
$$\vec{V}_S = \vec{A} \vec{V}_R + \vec{B} \vec{I}_R$$

At no load,
$$\vec{I}_R = 0$$

$\therefore$
$$\vec{V}_S = \vec{A} \vec{V}_{R0}$$

where
$$\vec{V}_{R0} = \text{voltage at receiving end at no load}$$

or
$$\vec{V}_{R0} = \vec{V}_S / \vec{A}$$

or
$$V_{R0} = V_S / A \text{ (in magnitude)}$$

$\therefore$
$$\% \text{ Regulation} = \frac{(V_S / A - V_R)}{V_R} \times 100 = \frac{(82.6 / 0.992) - 76.21}{76.21} \times 100 = 9.25\%$$

Equivalent circuit of LTL

Assumptions:

1. The line constants are uniformly distributed over the entire length.
2. The resistance and inductive reactance are the series elements.
3. The leakage susceptance (B) and leakage conductance (G) are shunt elements.
4. The leakage current through shunt admittance is maximum at the sending end.

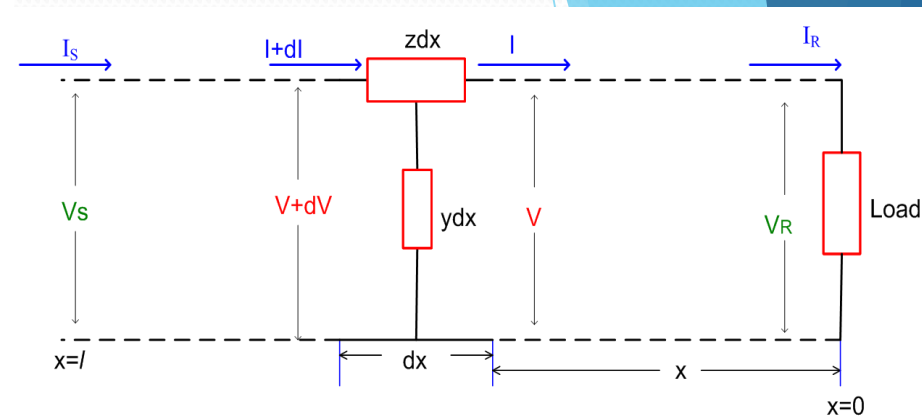


Fig. Equivalent circuit of long transmission line

- z = series impedance per unit length;
- y = shunt admittance per unit length;
- V = voltage at the end of element towards receiving end;
- $V + dV$ = voltage at the end of element towards sending end;
- $I + dI$ = current entering the element dx ;
- I = current leaving the element dx ;
- zdx = series impedance for small element dx ;
- ydx = shunt admittance for small element dx ;
- Y = Total admittance of the line
- Z = Total impedance of the line
- l = Total length

$$V_S = V_R \cosh(l\sqrt{yZ}) + \sqrt{\frac{Z}{y}} I_R \sinh(l\sqrt{yZ})$$

$$I_S = \sqrt{\frac{y}{Z}} V_R \sinh(l\sqrt{yZ}) + I_R \cosh(l\sqrt{yZ})$$

$$V_S = V_R \cosh \sqrt{Y Z} + I_R \sqrt{\frac{Z}{Y}} \sinh \sqrt{Y Z}$$

$$I_S = V_R \sqrt{\frac{Y}{Z}} \sinh \sqrt{Y Z} + I_R \cosh \sqrt{Y Z}$$

Detailed derivation of V and I can be found from Mehta, p 252.

Equivalent circuit of LTL

(iv) **Long lines—Rigorous method.** By rigorous method, the sending end voltage and current of a long transmission line are given by :

$$V_S = V_R \cosh \sqrt{Y Z} + I_R \sqrt{\frac{Z}{Y}} \sinh \sqrt{Y Z}$$

$$I_S = V_R \sqrt{\frac{Y}{Z}} \sinh \sqrt{Y Z} + I_R \cosh \sqrt{Y Z}$$

Comparing these equations with those of (i) and (ii), we get,

$$\vec{A} = \vec{D} = \cosh \sqrt{Y Z} ; \quad \vec{B} = \sqrt{\frac{Z}{Y}} \sinh \sqrt{Y Z} ; \quad \vec{C} = \sqrt{\frac{Y}{Z}} \sinh \sqrt{Y Z}$$

Incidentally

$$\begin{aligned} \vec{A} \vec{D} - \vec{B} \vec{C} &= \cosh \sqrt{Y Z} \times \cosh \sqrt{Y Z} - \sqrt{\frac{Z}{Y}} \sinh \sqrt{Y Z} \times \sqrt{\frac{Y}{Z}} \sinh \sqrt{Y Z} \\ &= \cosh^2 \sqrt{Y Z} - \sinh^2 \sqrt{Y Z} = 1 \end{aligned}$$

Problems of LTL

Example 10.15. A 3- ϕ transmission line 200 km long has the following constants :

$$\text{Resistance/phase/km} = 0.16 \, \Omega$$

$$\text{Reactance/phase/km} = 0.25 \, \Omega$$

$$\text{Shunt admittance/phase/km} = 1.5 \times 10^{-6} \, S$$

Calculate by rigorous method the sending end voltage and current when the line is delivering a load of 20 MW at 0.8 p.f. lagging. The receiving end voltage is kept constant at 110 kV.

Solution :

$$\text{Total resistance/phase,} \quad R = 0.16 \times 200 = 32 \, \Omega$$

$$\text{Total reactance/phase,} \quad X_L = 0.25 \times 200 = 50 \, \Omega$$

$$\text{Total shunt admittance/phase,} \quad Y = j 1.5 \times 10^{-6} \times 200 = 0.0003 \angle 90^\circ$$

$$\text{Series Impedance/phase,} \quad Z = R + j X_L = 32 + j 50 = 59.4 \angle 58^\circ$$

The sending end voltage V_S per phase is given by :

$$V_S = V_R \cosh \sqrt{Y Z} + I_R \sqrt{\frac{Z}{Y}} \sinh \sqrt{Z Y} \quad \dots(i)$$

$$\text{Now} \quad \sqrt{Z Y} = \sqrt{59.4 \angle 58^\circ \times 0.0003 \angle 90^\circ} = 0.133 \angle 74^\circ$$

$$Z Y = 0.0178 \angle 148^\circ$$

$$Z^2 Y^2 = 0.00032 \angle 296^\circ$$

Problems of LTL

$$\sqrt{\frac{Z}{Y}} = \sqrt{\frac{59.4 \angle 58^\circ}{0.0003 \angle 90^\circ}} = 445 \angle -16^\circ$$

$$\sqrt{\frac{Y}{Z}} = \sqrt{\frac{0.0003 \angle 90^\circ}{59.4 \angle 58^\circ}} = 0.00224 \angle 16^\circ$$

$$\begin{aligned} \cosh \sqrt{Y Z} &= 1 + \frac{Z Y}{2} + \frac{Z^2 Y^2}{24} \text{ approximately} \\ &= 1 + \frac{0.0178}{2} \angle 148^\circ + \frac{0.00032}{24} \angle 296^\circ \\ &= 1 + 0.0089 \angle 148^\circ + 0.0000133 \angle 296^\circ \\ &= 1 + 0.0089 (-0.848 + j 0.529) + 0.0000133 (0.438 - j 0.9) \\ &= 0.992 + j 0.00469 = 0.992 \angle 0.26^\circ \end{aligned}$$

$$\begin{aligned} \sinh \sqrt{Y Z} &= \sqrt{Y Z} + \frac{(Y Z)^{3/2}}{6} \text{ approximately} \\ &= 0.133 \angle 74^\circ + \frac{0.0024 \angle 222^\circ}{6} \\ &= 0.133 \angle 74^\circ + 0.0004 \angle 222^\circ \\ &= 0.133 (0.275 + j 0.961) + 0.0004 (-0.743 - j 0.67) \\ &= 0.0362 + j 0.1275 = 0.1325 \angle 74.6^\circ \end{aligned}$$

Problems of LTL

Receiving end voltage per phase is

$$V_R = 110 \times 10^3 / \sqrt{3} = 63508 \text{ V}$$

Receiving end current,
$$I_R = \frac{20 \times 10^6}{\sqrt{3} \times 110 \times 10^3 \times 0.8} = 131 \text{ A}$$

Putting the various values in exp (i), we get,

$$\begin{aligned} V_S &= 63508 \times 0.992 \angle 0.26^\circ + 131 \times 445 \angle -16^\circ 0' \times 0.1325 \angle 74^\circ 6' \\ &= 63000 \angle 0.26^\circ + 7724 \angle 58^\circ 6' \\ &= 63000 (0.999 + j 0.0045) + 7724 (0.5284 + j 0.8489) \\ &= 67018 + j 6840 = 67366 \angle 5^\circ 50' \text{ V} \end{aligned}$$

Sending end line-to-line voltage $= 67366 \times \sqrt{3} = 116.67 \times 10^3 \text{ V} = \mathbf{116.67 \text{ kV}}$

The sending end current I_S is given by :

$$I_S = V_R \sqrt{\frac{Y}{Z}} \sinh \sqrt{Y Z} + I_R \cosh \sqrt{Y Z}$$

Putting the various values, we get,

$$\begin{aligned} I_S &= 63508 \times 0.00224 \angle 16^\circ \times 0.1325 \angle 74^\circ 6' + 131 \times 0.992 \angle 0.26^\circ \\ &= 18.85 \angle 90^\circ 6' + 130 \angle 0.26^\circ \\ &= 18.85 (-0.0017 + j 0.999) + 130 (0.999 + j 0.0045) \\ &= 129.83 + j 19.42 = 131.1 \angle 8^\circ \text{ A} \end{aligned}$$

$\therefore$ Sending end current = **131.1 A**

Please check Tutorial problems 1 and 2 on page 255, Mehta.

Reference: Principles of Power System, V.K. Mehta and Rohit Mehta, S. Chand & Company.



QUESTIONS AND DISCUSSIONS

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